

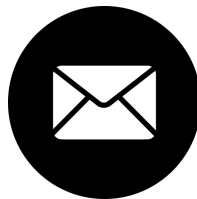


Blood Factory

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INSTALLATION

Requirements

- Minimum Unity Version 2022 LTS +
- URP or HDRP (Built-In Render Pipeline is not supported)

Installation

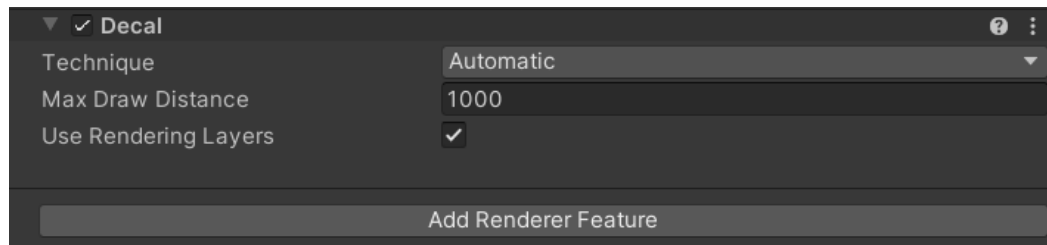
If you have other tools from Pampel Games installed in your project, it is recommended that you update them to the latest version before importing Road Constructor.

After importing, the 'PampelGames' folder can be moved to a different location within the 'Assets/' folder if desired.

URP

The included shaders are configured by default for the Universal Render Pipeline. However, some manual steps are required to ensure the decals work properly:

1. Select the active Universal Renderer Data asset in your project. By default, it is located in Assets/Settings.
2. In the Inspector, add the 'Decal' Renderer Feature and ensure that 'Use Rendering Layers' is checked.



And that's it!

For instructions on how to set up decal layers, please refer to the [Decal Layers](#) section.

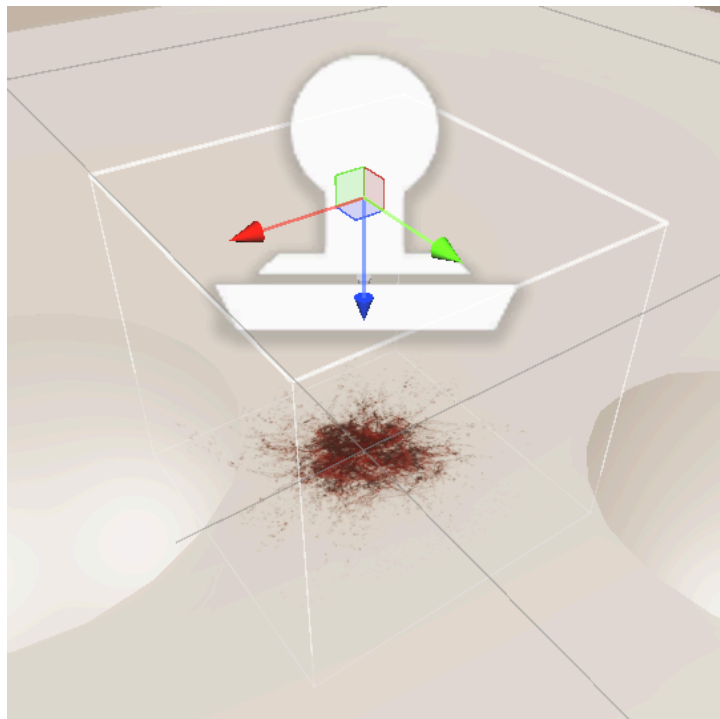
HDRP

The included shaders are set up by default for the Universal Render Pipeline.

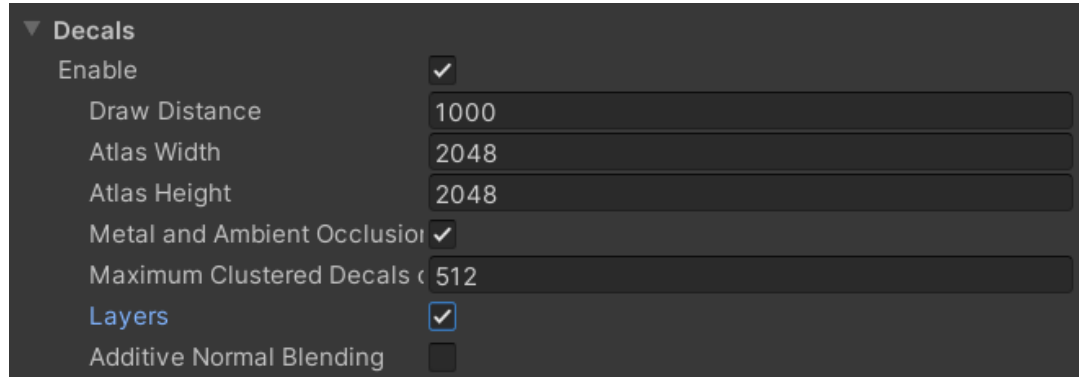
If you are using the High Definition Render Pipeline (HDRP), you need to import the HDRP shader package. This package is located in the Assets/PampelGames/BloodFactory folder, named 'BloodFactory_HDRP'.

Additionally, some manual steps are required to ensure the decals work properly. By default, decals are drawn onto any object, but you may want to restrict them to specific decal layers.

1. First, open a new scene and add a ground plane, or simply open the demo scene located in Assets/PampelGames/BloodFactory/Demo.
2. Drag one of the included decals into the scene and position it slightly above the ground, with the z-axis facing downward. The decals can be found in Assets/PampelGames/BloodFactory/Content/Assets/Decals. This step is simply to ensure that everything is working as intended.

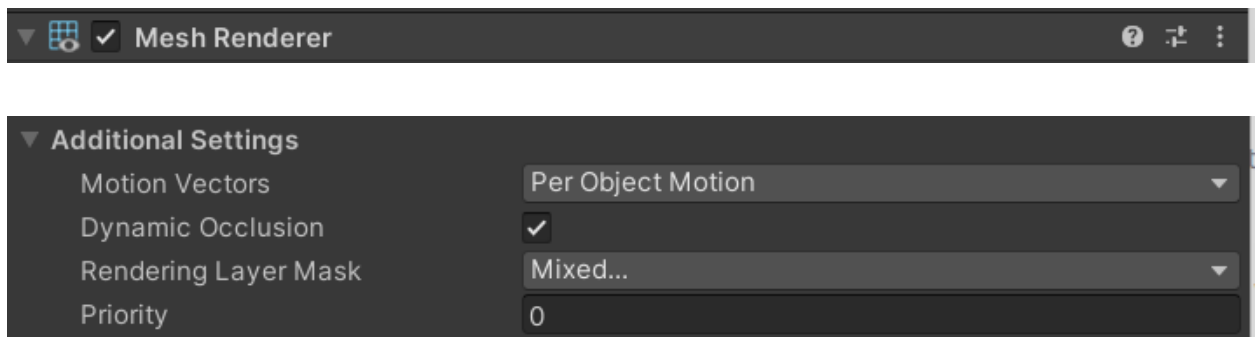


3. Navigate to the active HD Render Pipeline Asset. By default, it is located in Assets/Settings.
4. Ensure that decals are enabled and the 'Layers' setting is activated. At this point, the decal in the scene will no longer appear on the ground, as the layers do not match.



5. To make the decals appear, set the layer of all objects where you want the decal to be drawn to match the layer specified in the HDRP Decal Projector component.

By default, the '0: Decal Layer Default' is used. You can assign this layer in the Rendering Layer Mask in the Mesh Renderer - Additional Settings on the ground objects in your scene.



QUICK START

Overview

First, ensure that the render pipeline you are using has been set up correctly according to the [installation instructions](#).

The included effects are located in Assets/PampelGames/BloodFactory/Content/Prefabs and are categorized as follows:

Blood Splash

Directional and radial effects, ideal for character hits and explosions.

Blood Splatter

Primarily designed for detached body parts, such as when cutting off an arm.

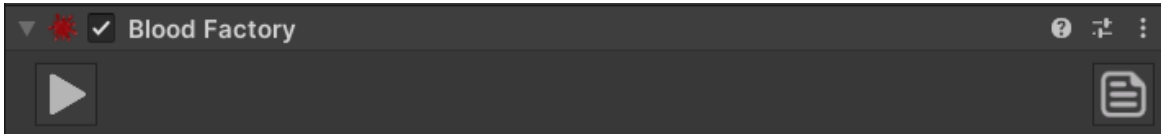
Mesh Decals

Can be used to animate flowing blood on walls.

Execution

To check out the effects, you have two options:

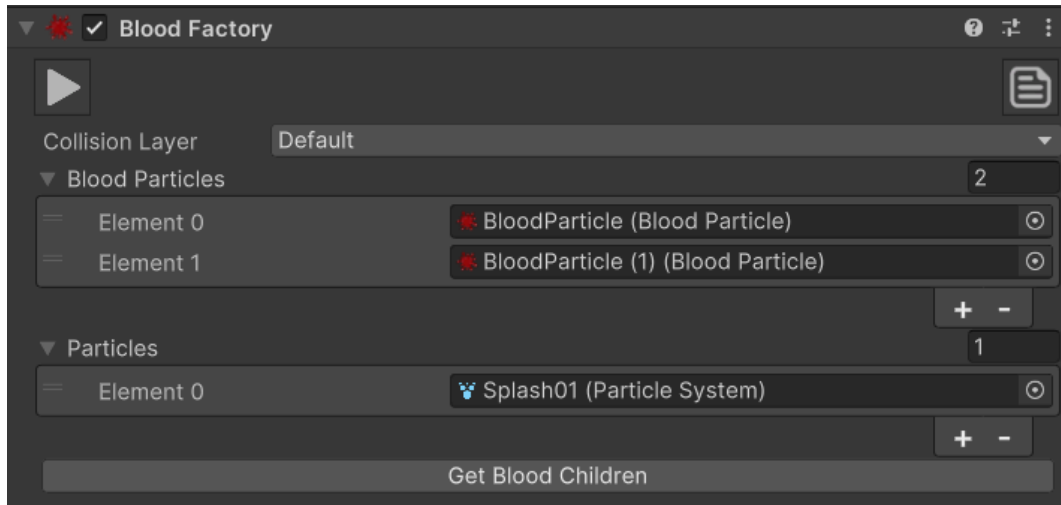
- 1) (Recommended) Open the demo scene located in Assets/PampelGames/BloodFactory/Demo, hit play, and use the UI buttons to execute.
- 2) Drag the effects into your scene and execute either by clicking the play icon at the top left of the component or by calling `bloodFactory.Execute()` manually (both runtime only).



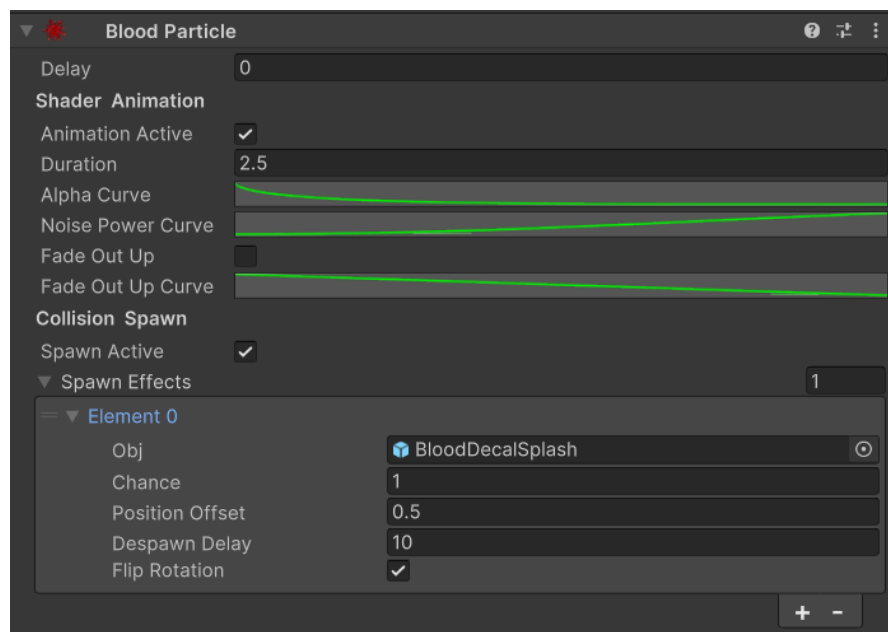
Customization

The effects are designed with easy extensibility and customization in mind.

Each effect is managed by its own BloodFactory component, which handles both the 'Blood Particle' and its particle effect children. For convenience, the collision layer specified in the BloodFactory component will override the layers of the referenced effects.



A 'Blood Particle' refers to a particle effect with an additional component that manages shader animations and object spawning on collision.



Particle effects in Blood Particle components typically have meshes referenced in the Particle System Renderer.



The meshes have specific UVs configured to change their appearance as the alpha maps fade in and out.

To visualize this behavior, drag a mesh into your scene (located in Assets/PampelGames/BloodFactory/Content/Assets/Meshes), locate the referenced material in the Mesh Renderer, and adjust the Alpha Strength, Noise Power, and Noise Strength values of the material.

INTEGRATIONS

Gore Simulator

Blood Factory integrates seamlessly with Gore Simulator.

Steps for Integration:

- 1) Ensure both Blood Factory and Gore Simulator are updated to their latest versions from the Asset Store.
- 2) Drag the effect prefabs into a Spawn Effects module for cuts or explosions. Be sure to reference them as GameObjects rather than Particle Effects to ensure proper execution.

Note: The collision layer of the Blood Factory component should differ from that of the character colliders if you want to spawn ground decals.

Effect Specialization:

- Blood Splashes: Ideal for hits and explosions.
- Blood Splatter Effects: Ideal for cut operations.

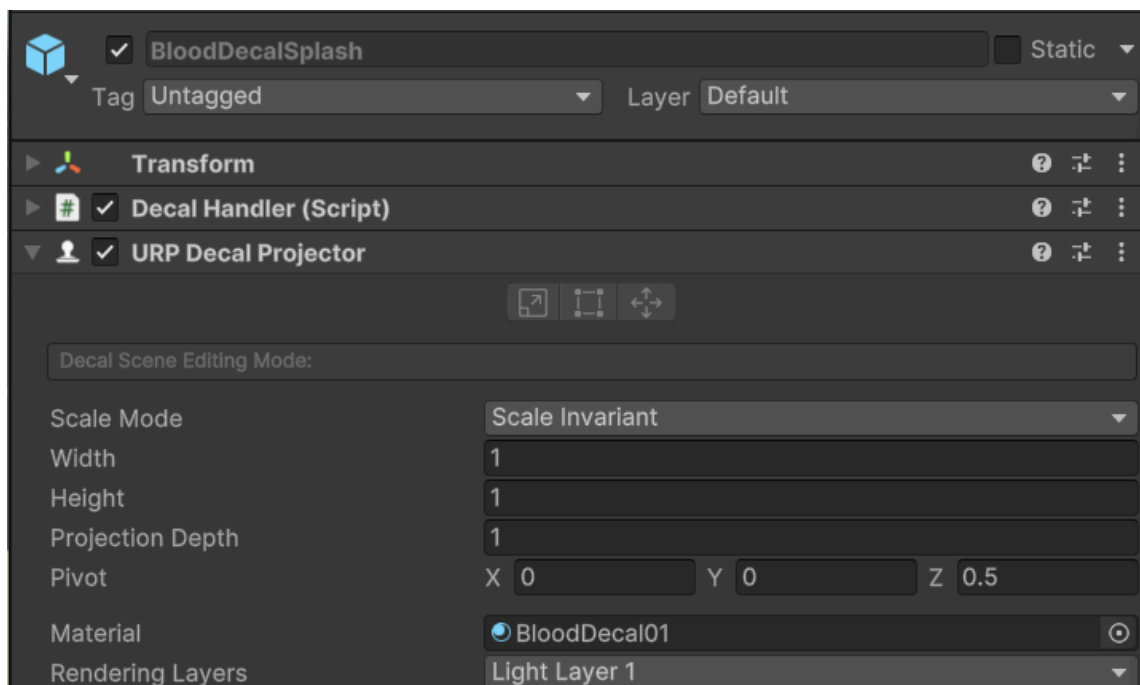
TROUBLESHOOTING

Decal Layers

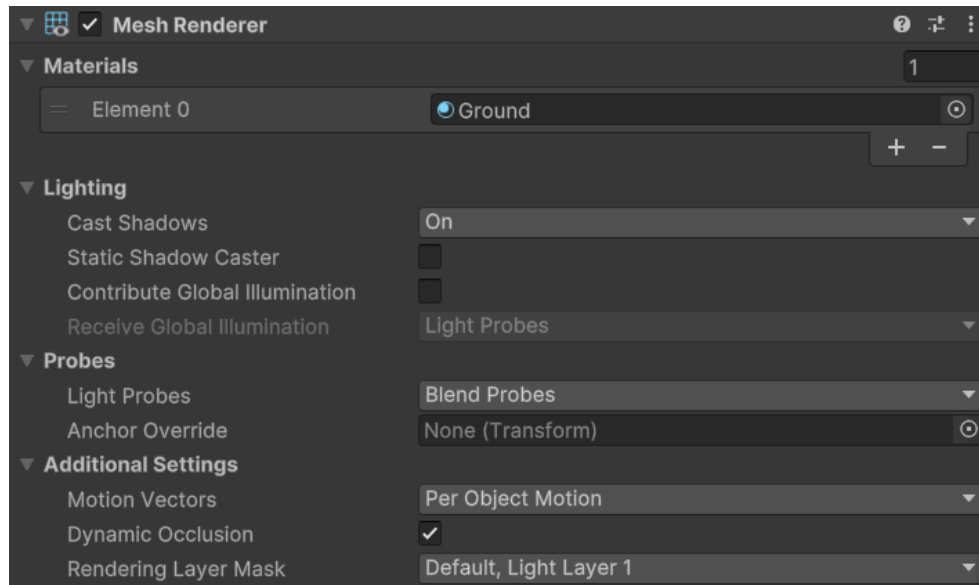
The included decals can be found in the Assets/PampelGames/Content/Decals folder.

By default, the decals are rendered only on the LightLayer1 Rendering Layer. In the demo scene, the ground objects are also configured with this layer; otherwise, the decals would not be visible.

The light layer is required; otherwise, the decals may spawn on characters or other moving objects, for example.

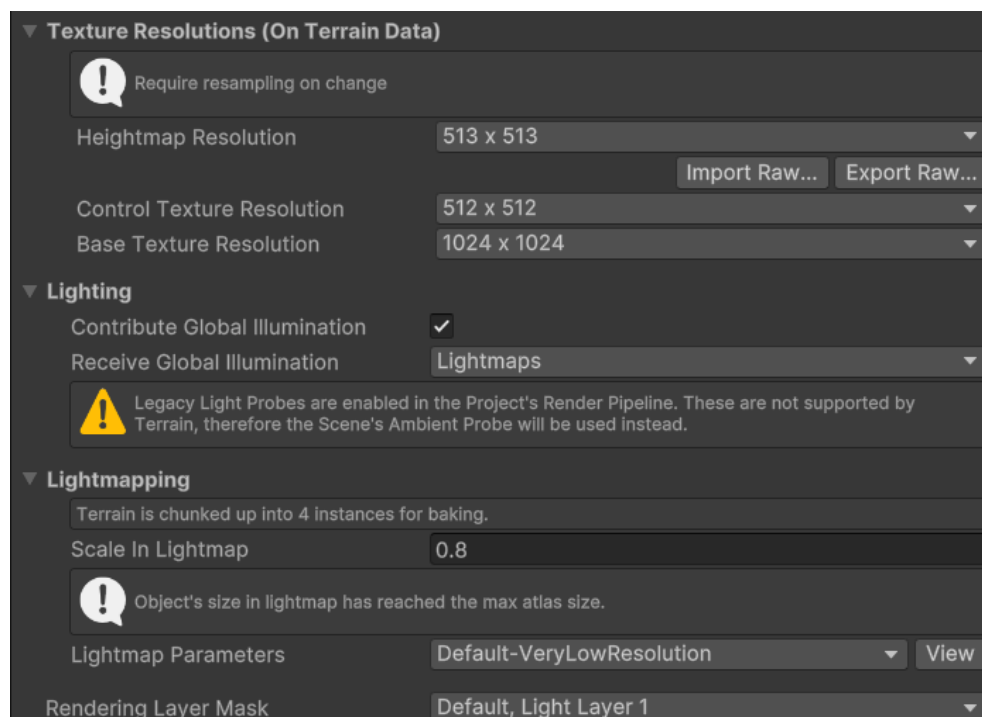


One of the decal prefabs, with Rendering Layers set to Light Layer 1



The Rendering Layer Mask setting can be found in the Mesh Renderer under Additional Settings

If you spawn decals on Unity terrains, the terrains must also have the decals' Rendering Layer Mask. You can find this setting under Terrain → Terrain Settings.

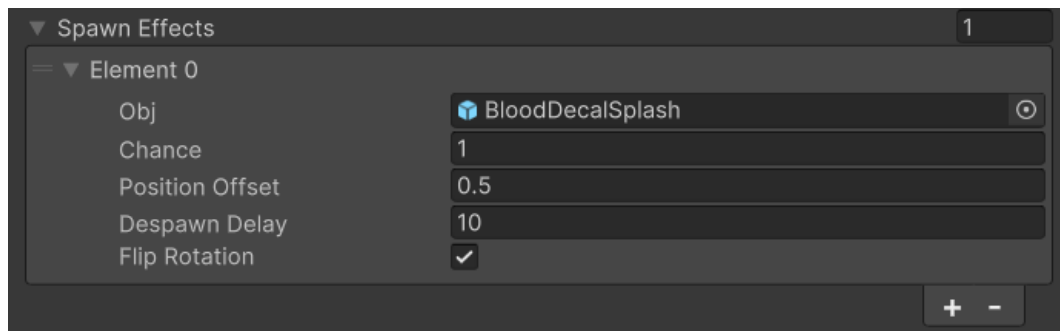


Screenshot of the Terrain Settings, with the Rendering Layer Mask found in the Lightmapping section.

Decal Performance

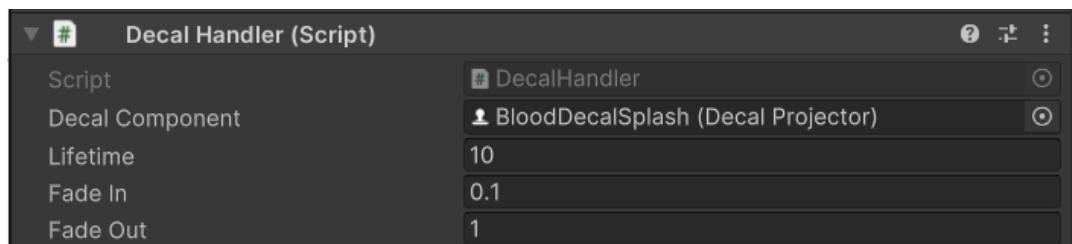
The effects are highly optimized for runtime use. If you're experiencing a bottleneck, it's most likely due to having too many unspawned decals active in the scene. The default effects can spawn dozens of decals, and if you stack or frequently trigger effects, you may end up with too many.

To address this issue, you have the following options, which can be found in the 'BloodParticle' component, a child of the 'BloodFactory' component:



- 1) Reduce the lifetime of the decals by adjusting the 'Despawn Delay'.

If you do that, make sure to adjust the lifetime of the decal prefabs as well to match the despawn timing.



- 2) Reduce the 'Chance' value to spawn fewer decals. If you find your scene isn't gory enough, you can increase the size of the Decal Handler.